

Table 1: Feature Description of Industrial Dataset. In the 'Comment', we specify which columns are utilized as features for training the model and which ones are excluded from the feature set (they are required for metrics & utilizing UCB-FE).

Feature ID	Feature Name	Feature Description	Comment
1	request_id	the unique identifier for a query	not trainable, q_t
2	item_id	the unique identifier for an item	not trainable, k
3	itemimps	weighted number of item's impressions based on position	not trainable, N_k
4	item_shows	number of item's shows	not trainable
5-8	bm_*	bm25 based	
9-14	mcat*	categories	categorical
15	cat_id	subcategory id	categorical
16-19	tfidf_*	tf-idf based	
20-30	idf_*	idf based	
31-36	trf_*	trigramm based	
37-40	query_info_*	query text characteristic	
41-45	item_info	item text characteristic	
46	same_loc	indicator user&item in the same location $\in \{0, 1\}$	
47	dist	the distance from the item to the user	
48	item_delivery	item's delivery indicator $\in \{0, 1\}$	
49	item_region_id	item region id	categorical
50	item_location_id	item location id	categorical
51-55	word_count_*	based on the number of unique request tokens	
56-58	hit_count_*	based on the non-unique token matches	
59-62	hit_pos_*	positional features based on matches	
63-65	len_*	based on length of text	
66-70	lcs_*	longest common sequence	
71-74	wlccs_*	weighted longest common contiguous subsequence	
75-78	lccs_*	longest common contiguous subsequence	
79-82	gaps_*		
83-85	atc_*	aggregate term closeness	
86-89	exact_order_*	based on matching the word order	
90-95	phrase_decay_*	based on phrase decay	
96-98	group_ctr_*	based on mean item's group ctr	
99	ctc	conversion (click to contact)	
100	ctr	aggregated click through rate	x^b
101	target	click indicator $\in \{0, 1\}$	target